SURFACES WITH CANONICAL MAP OF DEGREE THREE AND UNBOUNDED GEOMETRIC GENUS

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Abstract. We construct minimal surfaces of general type with arbitrarily large geometric genus whose canonical map has degree three onto a rational surface. Thus we answer negatively, for degree three, a question of Mendes Lopes and Pardini on the boundedness of the invariants of surfaces whose canonical map has odd degree onto a surface of geometric genus zero.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

Let X be a minimal surface of general type. The canonical map of X is the rational map

$$\varphi_{|K_X|} : X \dashrightarrow \mathbb{P}^{p_g(X)-1}$$

defined by $|K_X|$, and the canonical image $\Sigma \subset \mathbb{P}^{p_g(X)-1}$ is the closure of the image of $\varphi_{|K_X|}$. When Σ is a surface, we let $d := \deg \varphi_{|K_X|}$ be the degree of $\varphi_{|K_X|}$ onto Σ, and we let $p_g(\Sigma)$ be the geometric genus of a resolution of Σ. Beauville [Be79, Theorem 3.1] (see [MP23, Theorem 2.1]) showed that, when Σ is a surface, one of the following holds:

(A) $p_g(\Sigma) = 0$;
(B) $p_g(\Sigma) = p_g(X)$, and Σ is the canonical image of a surface of general type whose canonical map is birational.

In case (A), d and the invariants of X satisfy

$$(d-9)(p_g(X)-2) \leq 27 - 9q(X)$$

[MP23, Section 2.2, inequality (1)]. Thus, as [Be79] already pointed out, $d \leq 9$ in case (A) when $p_g(X) \geq 30$ [MP23, Section 2.2]. Xiao [Xia86, Theorem 3] proved that $d \leq 8$ in case (A) when $p_g(X) > 132$ (see [MP23, Section 2.2]). In case (A), families with unbounded $p_g(X)$ are known for any even $d \leq 8$ [Be79, Exemple 4.3], [MP23, Section 5.1]. Since $d = 1$ implies $p_g(\Sigma) = p_g(X) > 0$, any odd d in case (A) with $p_g(X) > 132$ satisfies $d \in \{3, 5, 7\}$.

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For $d = 3$, Mendes Lopes and Pardini [MP00, Theorem 1.1(a)] proved that, if $p_g(X) \geq 4$, the general member of $|K_X|$ is smooth, and Σ is a surface of minimal degree in $\mathbb{P}^{p_g(X)-1}$, that is,

$$\deg \Sigma = p_g(X) - 2,$$

then Σ is a cone and $p_g(X) \leq 5$. For $d \in \{3, 5, 7\}$, only sporadic examples are known [MP23, Section 5.1], and Mendes Lopes and Pardini asked the following question.

Question 1.1 ([MP23, Question 5.1])**.** Let $d \in \{3, 5, 7\}$. Consider the minimal surfaces X of general type for which $\varphi_{|K_X|}$ is generically finite of degree d onto a surface Σ with $p_g(\Sigma) = 0$. Are the invariants of all such X bounded?

Mendes Lopes, Pardini, and Pignatelli considered minimal surfaces X of general type such that $p_g(X) \geq 4$, $\varphi_{|K_X|}$ is generically finite, and

(a) the general member of $|K_X|$ is smooth;
(b) d is odd;
(c) Σ is ruled by lines, that is, through a general point of Σ there exists a line of $\mathbb{P}^{p_g(X)-1}$ contained in Σ.

For such X, they proved that

$$p_g(X) \leq d + 2,$$

and that Σ is the cone over the rational normal curve of degree $p_g(X)-2$ [MPP26, Theorem 3.1(i)]. Without the assumptions $p_g(X) \geq 4$ and (c), they also proved that $p_g(X) \leq 111$ if $d = 7$ [MPP26, Corollary 3.3]. In [MPP26, Section 1], they wrote that it is an open question whether the invariants of minimal surfaces X of general type with $p_g(X) \geq 4$ and with $\varphi_{|K_X|}$ generically finite of odd degree $d > 1$ are bounded, and that their results seem to point to a positive answer.

The goal of this paper is to answer Question 1.1 for $d = 3$.

Main result. Our main result is the following.

Theorem 1.2. *Let $n \geq 3$ be an integer. Then there exists a smooth minimal projective surface X of general type satisfying the following.*

(1) $p_g(X) = 2n - 2$.
(2) The canonical map $\varphi_{|K_X|} : X \dashrightarrow \mathbb{P}^{2n-3}$ is generically finite of degree 3 onto the canonical image Σ.
(3) Σ is projectively equivalent to the image of the embedding $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^{2n-3}$ defined by $|\mathcal{O}_{\mathbb{P}^1\times\mathbb{P}^1}(n-2, 1)|$.

Since $\Sigma \cong \mathbb{P}^1 \times \mathbb{P}^1$ is rational, $p_g(\Sigma) = 0$. Thus, for any integer $n \geq 3$, the surface X in Theorem 1.2 satisfies the hypotheses of Question 1.1 with $d = 3$, and

$$p_g(X) = 2n - 2 \to +\infty \quad \text{as } n \to +\infty.$$

Hence the answer to Question 1.1 is negative for $d = 3$. In particular, the answer to the question in [MPP26, Section 1] is also negative. Question 1.1 remains open for $d = 5$ and $d = 7$.

Remark 1.3. Let X be the surface in Theorem 1.2. For any $x \in \mathbb{P}^1$, the image of $\{x\} \times \mathbb{P}^1$ under the embedding in Theorem 1.2(3) is a line, so Σ is ruled by lines. Since Σ is smooth, it is not a cone, so a general member of $|K_X|$ is singular by [MPP26, Theorem 3.1(i)], and hypothesis (a) cannot be removed from [MPP26, Theorem 3.1(i)]. In particular, $|K_X|$ has base points, which also follows from [GP03, Theorem 4.2].

Sketch of the proof. Let Y be the ruled surface of Set-up 2.1 over a smooth projective curve of genus 1, with the section C_0, a fiber F, and the point y with $C_0 \cap F = \{y\}$. Resolving the base points of a general pencil $\Lambda \subset |3C_0 + F|$ by $g : Y' \to Y$, we obtain a fibration $h : Y' \to \Lambda \cong \mathbb{P}^1$ whose fiber over a general member $D \in \Lambda$ is $D' = g_*^{-1}D \cong D$, a smooth connected curve of genus 3 (Construction 3.1, Proposition 3.2(1), and Theorem 2.2(4)). For $n \geq 3$, the surface X in Theorem 1.2 is the minimal model of the base change $X' = Y' \times_\Lambda \mathbb{P}^1$ of h by $t = u^n$

(Construction 4.1), where t is a coordinate on Λ whose zero and pole are general members D_0 and D_∞ of Λ, so that X' is smooth (Lemma 4.2). Let $\psi = s_1/s_0$ for a basis s_0, s_1 of $H^0(Y, \mathcal{O}_Y(C_0 + F))$ (Construction 3.1(4)), and let ι be the embedding of $\mathbb{P}^1 \times \mathbb{P}^1$ into $\mathbb{P}^{2n-3}$ defined by $|\mathcal{O}_{\mathbb{P}^1\times\mathbb{P}^1}(n-2,1)|$. With the projections $f : X' \to Y'$ and $h' : X' \to \mathbb{P}^1$, Theorem 1.2 follows from

$$\varphi_{|K_{X'}|} = \iota \circ (u, f^*g^*\psi) : X' \dashrightarrow \mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^{2n-3}, \qquad \deg(u, f^*g^*\psi) = 3$$

(Section 5).

By Proposition 4.3, the 2-forms $u^{-j}f^*\eta_i$, $i \in \{0,1\}$, $1 \le j \le n-1$, form a basis of $H^0(X', \omega_{X'})$, where η_0, η_1 correspond to g^*s_0, g^*s_1 under an isomorphism $\omega_{Y'}(D'_\infty) \cong g^*\mathcal{O}_Y(C_0 + F)$ (Construction 4.1 and Proposition 3.2(2)). Since $\eta_1/\eta_0 = g^*\psi$, the formula for $\varphi_{|K_{X'}|}$ follows (Section 5). Moreover, $\deg(u, f^*g^*\psi) = \deg \psi|_D = 3$ for a general $D \in \Lambda$, since $g \circ f$ maps a general fiber of h' isomorphically onto such a D (Proposition 3.2(1) and Lemma 5.1).

Lemma 5.1 holds since, for a general $D \in \Lambda$, $s_0|_D, s_1|_D$ span a pencil in $H^0(D, \omega_D)$, $\deg \omega_D = 4$, whose only base point is y (Theorem 2.2(3)), with multiplicity 1 since $h^0(D, \omega_D(-2y)) = h^0(Y, \mathcal{O}_Y(2C_0)) = 1$ (Theorem 2.2(1)). Thus $\deg \psi|_D = 4 - 1 = 3$.

Structure of the paper. In Section 2 we fix an indecomposable ruled surface over an elliptic curve, the projectivization of the non-split extension of the structure sheaf of the elliptic curve by itself, and recall from [Ser23] the computations of linear systems on it used later. In Section 3 we construct a fibration of genus 3 on a blow-up of the ruled surface of Section 2. In Section 4 we construct cyclic base changes of the fibration of Section 3 and compute all canonical forms of the surfaces obtained. In Section 5 we show that a rational function on the ruled surface restricts to a morphism of degree 3 from a general fiber of the fibration to $\mathbb{P}^1$, and prove the main result.

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2. Preliminaries

Set-up 2.1. Let T, $\mathcal{V}$, Y, π, C_0, F, y be as follows.

(1) T is a smooth projective curve of genus 1.
(2) $\mathcal{V}$ is the unique, up to isomorphism, locally free sheaf of rank 2 on T that fits into a non-split exact sequence
$$0 \to \mathcal{O}_T \to \mathcal{V} \to \mathcal{O}_T \to 0.$$
(3) $Y := \mathbb{P}(\mathcal{V})$, and $\pi : Y \to T$ is the projection.
(4) $C_0 \subset Y$ is the section of π with $\mathcal{O}_Y(C_0) \cong \mathcal{O}_Y(1)$.
(5) F is a fiber of π, and $y \in Y$ is the point with $C_0 \cap F = \{y\}$.

Theorem 2.2 ([Ser23, Lemma 2.2(i)–(iv)])**.** *Notation as in Set-up 2.1. Then, for any integer $m \ge 1$, the following hold.*

(1) $h^0(Y, \mathcal{O}_Y(mC_0)) = 1$.
(2) $h^0(Y, \mathcal{O}_Y(mC_0 + F)) = m + 1$.
(3) $\mathrm{Bs}|mC_0 + F| = \{y\}$.
(4) A general member of $|mC_0 + F|$ is a smooth connected curve of genus m.

3. A genus three fibration

Construction 3.1. Notation as in Set-up 2.1.

(1) Let $\Lambda \subset |3C_0 + F|$ be a general pencil, and let $D_0, D_\infty \in \Lambda$ be distinct general members.
(2) Let $g : Y' \to Y$ be the minimal resolution of the base points of Λ, so that Λ induces a morphism $h : Y' \to \Lambda$. For any $D \in \Lambda$, let $D' := g_*^{-1}D$.

(3) Let t be a coordinate on $\Lambda \cong \mathbb{P}^1$ with zero D_0 and pole D_∞.
(4) Let s_0, s_1 be a basis of $H^0(Y, \mathcal{O}_Y(C_0 + F))$, which is 2-dimensional by Theorem 2.2(2) with $m = 1$, and let ψ be the rational function s_1/s_0 on Y.

By Theorem 2.2(4) with $m = 3$, a general member of Λ, in particular D_0 and D_∞, is a smooth connected curve of genus 3.

Proposition 3.2. *Notation as in Construction 3.1. Then the following hold.*

(1) For a general member $D \in \Lambda$, the fiber of h over the point D of Λ is D', and g restricts to an isomorphism $D' \to D$. In particular, $\operatorname{div}(h^*t) = D'_0 - D'_\infty$.
(2) $K_{Y'} + D'_\infty \sim g^*(C_0 + F)$.

Proof. Write g as a composite of blow-ups

$$Y' = Y_N \to Y_{N-1} \to \cdots \to Y_0 = Y,$$

where $b_i : Y_i \to Y_{i-1}$ is the blow-up of a base point of $(g_{i-1})_*^{-1}\Lambda$ for any $1 \le i \le N$, and $g_i := b_1 \circ \cdots \circ b_i : Y_i \to Y$ for any $0 \le i \le N$ (so $g_0 = \mathrm{id}_Y$ and $g_N = g$). For a smooth member $D \in \Lambda$, let $D_{Y_i} := (g_i)_*^{-1}D$. By induction on i, since $D_{Y_{i-1}}$ is smooth and the center of b_i is a base point of $(g_{i-1})_*^{-1}\Lambda$, $D_{Y_{i-1}}$ passes through the center of b_i with multiplicity 1, so D_{Y_i} is smooth and $g_i|_{D_{Y_i}} : D_{Y_i} \to D$ is an isomorphism.

(1) By the multiplicity 1 above, there exists a g-exceptional divisor Γ with $g^*D - D' = \Gamma$ for any smooth member $D \in \Lambda$. Since D'_0 and D'_∞ have no common component, Γ is the fixed part of $g^*\Lambda$, and h is defined by the base point free pencil $g^*\Lambda - \Gamma$. Thus for a general member $D \in \Lambda$, the fiber of h over D is $g^*D - \Gamma = D'$, and $g|_{D'} : D' \to D$ is an isomorphism. Since D_0 and D_∞ are general members, the zero divisor and the polar divisor of h^*t are D'_0 and D'_∞ respectively, so

$$\operatorname{div}(h^*t) = D'_0 - D'_\infty.$$

(2) Let $D = D_\infty$ and fix $1 \le i \le N$. Let E_i be the exceptional curve of b_i. Since $D_{Y_{i-1}}$ passes through the center of b_i with multiplicity 1, we have $K_{Y_i} \sim b_i^* K_{Y_{i-1}} + E_i$ and $D_{Y_i} = b_i^* D_{Y_{i-1}} - E_i$. Thus $K_{Y_i} + D_{Y_i} \sim b_i^*(K_{Y_{i-1}} + D_{Y_{i-1}})$. By induction on i,

$$K_{Y'} + D'_\infty \sim g^*(K_Y + D_\infty) \sim g^*(-2C_0 + 3C_0 + F) = g^*(C_0 + F).$$

The proposition follows. □

4. Canonical forms of cyclic base changes

Construction 4.1. Notation as in Construction 3.1. Let $n \ge 1$ be an integer.

(1) Let u be a coordinate on $\mathbb{P}^1$, let $X' := Y' \times_\Lambda \mathbb{P}^1$ be the fiber product of h and the morphism $\mathbb{P}^1 \to \Lambda$ given by $t = u^n$, and let $f : X' \to Y'$ and $h' : X' \to \mathbb{P}^1$ be the projections. We also write u for h'^*u.
(2) Let $\widetilde{D}_0$ and $\widetilde{D}_\infty$ be the reduced fibers of h' over 0 and ∞, respectively.
(3) Fix an isomorphism

$$g^*\mathcal{O}_Y(C_0 + F) \cong \omega_{Y'}(D'_\infty)$$

(Proposition 3.2(2)), and let η_0, η_1 be the images of g^*s_0, g^*s_1 in $H^0(Y', \omega_{Y'}(D'_\infty))$. We regard $H^0(Y', \omega_{Y'}(D'_\infty))$ as the space of rational 2-forms β on Y' with $\operatorname{div}(\beta) + D'_\infty \ge 0$, so that η_0, η_1 are rational 2-forms.

Lemma 4.2. *Notation as in Construction 4.1. Then X' is an integral smooth projective surface, f is a finite flat morphism of degree n, and*

$$f^*D'_0 = n\widetilde{D}_0, \quad f^*D'_\infty = n\widetilde{D}_\infty.$$

Proof. Since D_0, D_∞ are general members of $|3C_0 + F|$, they are smooth connected curves (Theorem 2.2(4) with $m = 3$). Thus, by Proposition 3.2(1), D'_0 and D'_∞ are smooth prime divisors, and they are the fibers of h over the points D_0 and D_∞ of Λ, with multiplicity 1. Hence h is smooth at any point of $D'_0 \cup D'_\infty$. Since h is proper, there exists an open neighborhood

$U \subset \Lambda$ of $\{D_0, D_\infty\}$ such that h is smooth over U, so h' is smooth over the preimage U' of U in $\mathbb{P}^1$, an open neighborhood of $\{0, \infty\}$ in $\mathbb{P}^1$. Since the morphism $\mathbb{P}^1 \to \Lambda$ given by $t = u^n$ is étale over $\Lambda \backslash \{D_0, D_\infty\}$, f is étale over $Y' \backslash (D'_0 \cup D'_\infty)$. Thus X' is smooth on $f^{-1}(Y' \backslash (D'_0 \cup D'_\infty))$ and on $h'^{-1}(U')$, and

$$f^{-1}(Y' \backslash (D'_0 \cup D'_\infty)) \cup h'^{-1}(U') = X'.$$

Hence X' is smooth. X' is a closed subscheme of $Y' \times \mathbb{P}^1$, hence projective. The morphism f is the base change of the finite flat morphism $\mathbb{P}^1 \to \Lambda$, $t = u^n$, of degree n, hence finite flat of degree n. In particular, $\dim X' = 2$ at any point.

Let $u_0 \in \mathbb{P}^1$ be a general point, and let $D \in \Lambda$ be the member $t = u_0^n$, which is general. Since h' is flat and proper and the fiber $h'^{-1}(u_0) \cong D'$ of h' is connected (Proposition 3.2(1) and Theorem 2.2(4) with $m = 3$), X' is connected, hence integral since X' is smooth.

For the scheme-theoretic fibers, f induces isomorphisms

$$h'^{-1}(0) \cong h^{-1}(D_0) = D'_0, \qquad h'^{-1}(\infty) \cong h^{-1}(D_\infty) = D'_\infty,$$

and D'_0, D'_∞ are reduced. Thus $\operatorname{div}(u) = \widetilde{D}_0 - \widetilde{D}_\infty$. Since $f^* h^* t = u^n$, we have

$$f^* D'_0 - f^* D'_\infty = \operatorname{div}(f^* h^* t) = \operatorname{div}(u^n) = n\widetilde{D}_0 - n\widetilde{D}_\infty.$$

Since $\operatorname{Supp} f^* D'_0 = \widetilde{D}_0$ and $\operatorname{Supp} f^* D'_\infty = \widetilde{D}_\infty$ are disjoint, the lemma follows. □

Proposition 4.3. *Notation as in Construction 4.1. Then the rational 2-forms*

$$u^{-j} f^* \eta_i, \quad i \in \{0, 1\},\ 1 \le j \le n-1,$$

form a basis of $H^0(X', \omega_{X'})$. In particular, $p_g(X') = 2n - 2$.

Proof. **Step 1.** Since X' is defined by $(u^{-1})^n = (h^* t)^{-1}$ over $Y' \backslash D'_0$, any rational 2-form α on X' can be written uniquely as

$$\alpha = \sum_{j=0}^{n-1} u^{-j} f^* \beta_j$$

with rational 2-forms β_j on Y', and the summands are the eigencomponents of α under $u \to \zeta u$, $\zeta \in \mu_n$. Thus α is regular if and only if $u^{-j} f^* \beta_j$ is regular for any $0 \le j \le n-1$.

Step 2. Let β be a rational 2-form on Y' and $0 \le j \le n-1$ an integer. We may assume that $\beta \ne 0$. We show that

$$\operatorname{mult}_{\widetilde{D}_0} \operatorname{div}(u^{-j} f^* \beta) = n \operatorname{mult}_{D'_0} \operatorname{div}(\beta) + n - 1 - j, \tag{4.1}$$

$$\operatorname{mult}_{\widetilde{D}_\infty} \operatorname{div}(u^{-j} f^* \beta) = n \operatorname{mult}_{D'_\infty} \operatorname{div}(\beta) + n - 1 + j. \tag{4.2}$$

Let $q \in D'_0$ be a general point and z a regular function near q such that $h^* t, z$ are local coordinates at q. Over a neighborhood of q, X' is defined by $u^n = h^* t$, so $u, f^* z$ are local coordinates on X' at any point of $f^{-1}(q)$, and $\widetilde{D}_0 = (u = 0)$ near $f^{-1}(q)$. Write

$$\beta = (h^* t)^{\operatorname{mult}_{D'_0} \operatorname{div}(\beta)} b d(h^* t) \wedge dz$$

near q, where b is a rational function which is a unit at the generic point of D'_0. Since $f^* h^* t = u^n$,

$$u^{-j} f^* \beta = n u^{n \operatorname{mult}_{D'_0} \operatorname{div}(\beta) + n - 1 - j} (f^* b) du \wedge d(f^* z),$$

and $f^* b$ is a unit at the generic point of $\widetilde{D}_0$. This implies (4.1). Let $q \in D'_\infty$ be a general point and z a regular function near q such that $1/h^* t, z$ are local coordinates at q. Over a neighborhood of q, X' is defined by $(u^{-1})^n = 1/h^* t$, so $u^{-1}, f^* z$ are local coordinates on X' at any point of $f^{-1}(q)$, and $\widetilde{D}_\infty = (u^{-1} = 0)$ near $f^{-1}(q)$. Write

$$\beta = (1/h^* t)^{\operatorname{mult}_{D'_\infty} \operatorname{div}(\beta)} b d(1/h^* t) \wedge dz$$

near q, where b is a rational function which is a unit at the generic point of D'_∞. Since $u^{-j} = (u^{-1})^j$ and $f^*(1/h^*t) = (u^{-1})^n$,

$$u^{-j} f^*\beta = n(u^{-1})^{n\,\mathrm{mult}_{D'_\infty}\,\mathrm{div}(\beta)+n-1+j}(f^*b)d(u^{-1}) \wedge d(f^*z),$$

and f^*b is a unit at the generic point of $\widetilde{D}_\infty$. This implies (4.2). At any prime divisor $P \subset X'$ other than $\widetilde{D}_0, \widetilde{D}_\infty$, f is étale at the generic point of P and u is a unit there, so

$$\mathrm{mult}_P \,\mathrm{div}(u^{-j} f^*\beta) = \mathrm{mult}_{f(P)} \,\mathrm{div}(\beta).$$

Since $0 \le n-1-j \le n-1$, the right-hand side of (4.1) is non-negative if and only if $\mathrm{mult}_{D'_0} \,\mathrm{div}(\beta) \ge 0$. If $j = 0$, then $n-1+j = n-1$, so the right-hand side of (4.2) is non-negative if and only if $\mathrm{mult}_{D'_\infty} \,\mathrm{div}(\beta) \ge 0$. If $1 \le j \le n-1$, then $n \le n-1+j \le 2n-2$, so the right-hand side of (4.2) is non-negative if and only if $\mathrm{mult}_{D'_\infty} \,\mathrm{div}(\beta) \ge -1$. Since f is surjective and X' is smooth, we deduce:

- $f^*\beta$ is regular if and only if $\mathrm{div}(\beta) \ge 0$;
- for $1 \le j \le n-1$, $u^{-j} f^*\beta$ is regular if and only if $\mathrm{div}(\beta) + D'_\infty \ge 0$.

Step 3. Let $\alpha = \sum_{j=0}^{n-1} u^{-j} f^*\beta_j$ be a rational 2-form on X' as in **Step 1**. By **Steps 1 and 2**, $\alpha \in H^0(X', \omega_{X'})$ if and only if $\beta_0 \in H^0(Y', \omega_{Y'})$ and $\beta_j \in H^0(Y', \omega_{Y'}(D'_\infty))$ for any $1 \le j \le n-1$. Since Y' is birational to the ruled surface Y, $H^0(Y', \omega_{Y'}) = 0$. By Proposition 3.2(2) and since $g_*\mathcal{O}_{Y'} = \mathcal{O}_Y$,

$$H^0(Y', \omega_{Y'}(D'_\infty)) \cong H^0(Y', g^*\mathcal{O}_Y(C_0+F)) \cong H^0(Y, \mathcal{O}_Y(C_0+F)),$$

and the composite isomorphism sends η_i to s_i for $i \in \{0,1\}$. Thus η_0, η_1 form a basis of $H^0(Y', \omega_{Y'}(D'_\infty))$ (Theorem 2.2(2) with $m = 1$). Since the expression of α in **Step 1** is unique and f^* is injective on rational 2-forms, any $\alpha \in H^0(X', \omega_{X'})$ can be written uniquely as

$$\alpha = \sum_{j=1}^{n-1} \sum_{i \in \{0,1\}} c_{ij} u^{-j} f^*\eta_i, \quad c_{ij} \in \mathbb{C},$$

and any such sum is regular. The proposition follows. □

5. Proof of Theorem 1.2

Lemma 5.1. *Notation as in Construction 3.1. Let $D \in \Lambda$ be a general member. Then $s_0|_D \neq 0$, and the rational function $\psi|_D := s_1|_D / s_0|_D$ on D defines a morphism $D \to \mathbb{P}^1$ of degree 3.*

Proof. By Theorem 2.2(4) with $m = 3$, D is a smooth connected curve of genus 3. Since $K_Y \sim -2C_0$, adjunction implies $\omega_D \cong \mathcal{O}_D(K_Y + D) \cong \mathcal{O}_D(C_0 + F)$, and

$$\deg \omega_D = (C_0+F)\cdot(3C_0+F) = 3C_0^2 + 4C_0 \cdot F + F^2 = 4.$$

The kernel of the restriction map $H^0(Y, \mathcal{O}_Y(C_0+F)) \to H^0(D, \omega_D)$ is

$$H^0(Y, \mathcal{O}_Y(C_0+F-D)) = H^0(Y, \mathcal{O}_Y(-2C_0)) = 0.$$

Thus $s_0|_D \neq 0$, and $s_0|_D, s_1|_D$ are linearly independent in $H^0(D, \omega_D)$.

By Theorem 2.2(3) with $m = 3$ and with $m = 1$, we have $y \in D$ and $\mathrm{Bs}|C_0 + F| = \{y\}$. Thus the common zero locus of $s_0|_D$ and $s_1|_D$ is $\{y\}$. Since D is irreducible and $D \not\sim C_0$, $C_0 \not\subset D$, and since $C_0 \cdot D = 1$ and $y \in C_0 \cap D$, we have $C_0|_D = y$ as divisors on D. Since $2C_0 - D \sim -C_0 - F$, there exists an exact sequence

$$0 \to \mathcal{O}_Y(-C_0-F) \to \mathcal{O}_Y(2C_0) \to \mathcal{O}_D(2y) \to 0.$$

Since $(-C_0-F)\cdot F = -1$ and all fibers of π are numerically equivalent to F, $R^i\pi_*\mathcal{O}_Y(-C_0-F) = 0$ for any i, so $H^i(Y, \mathcal{O}_Y(-C_0-F)) = 0$ for any i. By Theorem 2.2(1) with $m = 2$,

$$h^0(D, \mathcal{O}_D(2y)) = h^0(Y, \mathcal{O}_Y(2C_0)) = 1.$$

Since D has genus 3, the Riemann–Roch theorem on D implies

$$h^0(D,\omega_D(-2y)) = h^0(D,\mathcal{O}_D(2y)) - (\deg(2y)+1-3) = 1.$$

Since $s_0|_D, s_1|_D$ are linearly independent in $H^0(D,\omega_D)$ and $h^0(D,\omega_D(-2y))=1$, $s_0|_D$ and $s_1|_D$ do not both vanish to order at least 2 at y, so the common zero divisor of $s_0|_D$ and $s_1|_D$ is exactly y. Therefore $s_0|_D, s_1|_D$ are sections of $\omega_D(-y)$ without common zeros, and

$$\deg \omega_D(-y) = 4-1 = 3.$$

Hence $\psi|_D$ defines a morphism $D\to\mathbb{P}^1$ of degree 3. The lemma follows. □

Proof of Theorem 1.2. Let $n\geq 3$ be an integer, and let X', u, f, h', η_0, η_1 be as in Construction 4.1. By Lemma 4.2, X' is an integral smooth projective surface. By Proposition 4.3, $p_g(X') = 2n-2$, and the canonical map $\varphi_{|K_{X'}|} : X' \dashrightarrow \mathbb{P}^{2n-3}$ is defined by the basis of Proposition 4.3. Since η_0,η_1 are the images of g^*s_0, g^*s_1 under one isomorphism of line bundles, $\eta_1/\eta_0 = g^*s_1/g^*s_0 = g^*\psi$. Thus for any $1\leq j\leq n-1$ we have

$$\frac{u^{-j}f^*\eta_0}{u^{-(n-1)}f^*\eta_0} = u^k \quad\text{and}\quad \frac{u^{-j}f^*\eta_1}{u^{-(n-1)}f^*\eta_0} = u^k f^*g^*\psi, \qquad k := n-1-j \in \{0,\dots,n-2\}. \tag{5.1}$$

Let $[x_0:x_1]$ be homogeneous coordinates on the $\mathbb{P}^1$ of Construction 4.1 such that $u = x_1/x_0$, and let $[y_0:y_1]$ be homogeneous coordinates on a second copy of $\mathbb{P}^1$. Let

$$(u, f^*g^*\psi) : X' \dashrightarrow \mathbb{P}^1\times\mathbb{P}^1$$

be the rational map whose first component is h' and whose second component is the rational map defined by $y_1/y_0 = f^*g^*\psi$. Let $\iota : \mathbb{P}^1\times\mathbb{P}^1 \to \mathbb{P}^{2n-3}$ be the morphism defined by the basis

$$x_0^{n-2-k}x_1^k y_i, \quad i\in\{0,1\},\ 0\leq k\leq n-2$$

of $H^0(\mathbb{P}^1\times\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1\times\mathbb{P}^1}(n-2,1))$. Dividing by $x_0^{n-2}y_0$, the coordinates of $\iota\circ(u,f^*g^*\psi)$ are u^k and $u^kf^*g^*\psi$, $0\leq k\leq n-2$. Comparing with (5.1),

$$\varphi_{|K_{X'}|} = \iota\circ(u,f^*g^*\psi).$$

Since $n-2\geq 1$, $\mathcal{O}_{\mathbb{P}^1\times\mathbb{P}^1}(n-2,1)$ is very ample, so ι is a closed embedding.

Let $u_0\in\mathbb{P}^1$ be a general point, and let $D\in\Lambda$ be the member $t=u_0^n$, which is general. By the definition of X' as $Y'\times_\Lambda\mathbb{P}^1$ and Proposition 3.2(1), $g\circ f$ restricts to an isomorphism $h'^{-1}(u_0)\to D$, under which the restriction of $(u,f^*g^*\psi)$ to $h'^{-1}(u_0)$ is $(u_0,\psi|_D)$. By Lemma 5.1, $\psi|_D : D\to\mathbb{P}^1$ is a morphism of degree 3. Thus $(u,f^*g^*\psi)$ maps a general fiber of h' by a morphism of degree 3 onto the corresponding fiber of the first projection $\mathbb{P}^1\times\mathbb{P}^1\to\mathbb{P}^1$, so it is dominant and generically finite of degree 3.

Therefore $\varphi_{|K_{X'}|}$ is generically finite of degree 3 onto the surface $\iota(\mathbb{P}^1\times\mathbb{P}^1)$. In particular, X' is of general type. Let X be the minimal model of X'. Then X is a smooth minimal projective surface of general type, $p_g(X) = p_g(X') = 2n-2$, and the composite of the contraction $X'\to X$ with $\varphi_{|K_X|}$ is $\varphi_{|K_{X'}|}$. Thus X satisfies (1)–(3). □

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